POSO CREEK – BAKERSFIELD, CALIFORNIA
Enhanced Oil Recovery (EOR) Program Report
RDV® – Dynamic Vasoactive Reactor

Context and Problem

The Poso Creek field is located in the San Joaquin Valley, near Bakersfield, California. It is a mature heavy oil field produced through cyclic steam injection. Wells operate with water cuts (BS&W) exceeding 90% and face natural production decline due to high viscosity and the accumulation of organic deposits (asphaltenes).

In 2022, E&B Natural Resources selected two wells for a pilot application with RDV-01L®:

Well	API Number	Operator
Conoco 3-5	04-029-91345	E&B Natural Resources
Conoco 6-10	04-029-91382	E&B Natural Resources

Baseline (2021):

- Conoco 3-5: 5.2 BOPD average
- Conoco 6-10: 9.5 BOPD average

Application Design

A matrix stimulation intervention was designed for Enhanced Oil Recovery (EOR) purposes using RDV® protonic chemical technology. A pill was prepared at the surface consisting of a 50% RDV® and 50% crude oil blend extracted from the same well.

Once the pill was placed above the wellbore fluid level, a 24-hour recirculation cycle was executed to ensure homogenization of the product with the crude oil present in the wellbore. Following the recirculation period, a soaking phase was initiated to allow the protonation reaction induced by RDV® to act upon the hydrocarbons in the homogenized fluid column, thereby initiating the molecular fragmentation process of long-chain hydrocarbons.

Subsequently, the treatment was displaced into the formation via forced injection (squeeze) using hot water as the drive fluid. This step extended the protonation effect into the porous rock with the following objectives:

- Clean the formation pores by removing organic deposits (asphaltenes) obstructing flow.

- Reduce crude oil viscosity, improving its mobility within the porous medium.
- Enhance relative permeability to oil, facilitating its extraction.

Protocol applied:

- Injection of RDV-01L® to the bottom of the well.
- Soaking time: 48–72 hours.
- Forced displacement into the formation (squeeze).
- Well opening for production.

Field Application

The application was carried out in June 2022. The product was injected into both wells following the established protocol, with no operational incidents. Production was continuously monitored through CalGEM and WellSTAR reports.

Results Obtained

A. Gross Production (BFPD)

Period	Conoco 3-5	Conoco 6-10
2021 (Baseline)	5.2 BOPD	9.5 BOPD
2022 (Post-RDV)	7.4 BOPD (+54%)	14.2 BOPD (+56%)
2023	7.1 BOPD	13.5 BOPD
2024	6.8 BOPD	12.9 BOPD
2025 – Feb 2026	6.5 BOPD	12.4 BOPD

B. Net Oil Production – Considering >90% BS&W

Indicator	Pre-RDV (2021)	Post-RDV (2022–2023)	Status (Feb 2026)
Water Cut (BS&W)	94% – 95%	88% – 91%	90% – 92%

Indicator	Pre-RDV (2021)	Post-RDV (2022–2023)	Status (Feb 2026)
Net Oil (BOPD)	5 – 6 bbl	10 – 13 bbl	8 – 9.5 bbl
Mobility Efficiency	Baseline	+50% Improvement	Sustained

C. Comparison with Offset Wells (Untreated)

In February 2026, 10 control wells in the Premier area, operating in the same formation but without RDV® treatment, were selected for comparison.

Category	Immediate Change (2022)	Cumulative Performance (2023–2024)	Decline (Feb 2026)
RDV®-Treated Wells	+53.5% (Increase)	+48% (Sustained)	-4% (Minimal)
Average of 10 Offset Wells	-6.2% (Decline)	-18.4% (Loss)	-29.5% (Critical)

D. Key Findings

1. **Decline curve reversal:** Treated wells reversed the natural decline trend (10–15% annually) and maintained production levels above baseline for over 40 months.
2. **Sustained effect:** The effect was not a momentary peak. Data from 2024 and 2025 confirm that the modification of crude mobility was durable.
3. **Fewer interventions:** Treated wells required 2.5 fewer mechanical cleaning interventions than control wells between 2023 and 2025.
4. **Synergy with thermal EOR:** Both wells were integrated into Phase III of E&B's Injection Project, confirming that RDV® successfully prepared the wells for more aggressive secondary/tertiary recovery methods with higher CAPEX.

E. Efficiency Gap

As of February 2026, the RDV®-treated wells maintained an **83% efficiency gap** over the average of offset wells.

F. Conclusions

The molecular fragmentation induced by RDV®, through the irreversible cleavage of C–C and C–H bonds in asphaltene chains, restored effective porosity in the formation and eliminated the organic constriction restricting the flow of the hydrocarbon phase toward the wells in the Poso Creek field. Consequently, relative permeability to oil, previously reduced by organic obstruction, improved significantly, re-establishing the productive capacity of the rock–fluid system.

The results evidenced that RDV® – Dynamic Vasoactive Reactor – contributed to sustaining porosity recovery over extended periods of productive activity. Four years after the initial intervention, the treated wells maintained production levels above both their baseline and the average production of surrounding untreated wells, confirming the persistence of the protonation effect.

The chemical protonation reaction, initiated during the initial RDV® application, remained active by relying exclusively on the natural energy of the reservoir – bottomhole pressure and temperature – as a catalyst, without requiring additional recirculation or product re-application. In this way, RDV® proved to be a long-term solution for mature heavy oil wells, with porosity constricted by organic deposits and water cuts exceeding 90%.

The protonic chemical technology demonstrated the following benefits:

- Increased net oil production by more than 50%.
- Sustained porosity recovery effect for over 4 years.
- Reduced frequency of mechanical interventions and other chemical treatments.
- Facilitated integration with thermal EOR methods in subsequent project phases.

Technical Department GQ – USA International LLC (Globalquimica)

Disclaimers

This document presents a summary of field results for informational and educational purposes only. The data and conclusions contained herein are based on specific operating conditions, well characteristics, and reservoir properties that may not be applicable to other fields or wells. GQ-USA International LLC (Globalquimica) makes no representations or warranties, express or implied, regarding the accuracy, completeness, or suitability of the information contained in this document for any particular purpose. All technical decisions should be made in consultation with qualified professionals and based on site-specific evaluations.

The product names and trademarks mentioned in this document are the property of their respective owners. Any reference to specific products or services does not constitute an endorsement or recommendation by GQ-USA International LLC.

The distribution, reproduction, or use of this document, in whole or in part, for commercial purposes without the prior written authorization of GQ-USA International LLC is strictly prohibited.